

QBITEL BRIDGE

BPO & CALL CENTER SECURITY PLATFORM

Quantum-Safe Security for the Human API

4–6 Hours

Full Deployment
Zero Downtime

78%

Autonomous Threat
Resolution

9 Frameworks

Compliance Automated

<2ms

Voice Encryption Overhead

89%+

Protocol Discovery Accuracy

\$10B+

Annual Industry Fraud Loss
QBITEL Prevents

AI-Powered • Quantum-Safe • Zero Disruption • Deployed in Hours

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1. THE PROBLEM BPOS CANNOT IGNORE

Three converging threats exposing contact center infrastructure

Business Process Outsourcing operations are the frontline of global commerce — processing millions of sensitive customer interactions daily. Credit card payments, patient records, insurance details, and financial disputes flow through infrastructure built decades ago, protected by encryption standards that were never designed for today's threat landscape.

■ **Quantum Harvest Attacks:** Nation-state adversaries capture encrypted call recordings today to decrypt when quantum computers arrive (5–10 years). SOX 7-year retention means recordings from this week will still exist when quantum decryption is viable.

■ **Toll Fraud — \$10B Annual Industry Loss:** A single compromised PBX trunk can generate \$50,000 in fraudulent premium-rate calls over one weekend. Traditional detection happens 72 hours later on the carrier invoice.

■ **Remote Workforce Security Gap:** 60%+ of BPO agents work from home on consumer-grade networks. One compromised home router is all a threat actor needs to intercept agent sessions and exfiltrate PII undetected.

■ *Traditional response: \$5M+ PBX replacement, 12–18 months of migration risk, weeks of agent retraining. QBITEL response: 4–6 hour network-layer deployment. Nothing replaced. Nothing disrupted.*

Challenge	Industry Impact	QBITEL Approach
Legacy PBX encryption	Unencrypted SIP carries 70%+ of global call traffic	PQC overlay — no hardware changes
TN3270e terminal sessions	Mainframe access with zero session protection	PQC tunnel wrapping — transparent
Payment DTMF exposure	50M+ daily voice payment transactions at risk	Automated DTMF masking in <5ms
Remote agent risk	60%+ agents on consumer-grade home networks	VPN-less PQC tunnels — instant
Toll fraud	\$10B+ annual industry loss	AI detection in <1 second
Insider data theft	34% of BPO breaches are insider threats	DLP enforcement at kernel level
Multi-tenant compliance	Separate audits per enterprise client	Isolated policies — one platform

2. QBITEL BRIDGE — HOW IT WORKS

AI discovers. PQC encrypts. Agentic AI defends.

QBITEL Bridge deploys as a network-layer security overlay in 4–6 hours. No PBX replacement. No agent retraining. No downtime. In three automated phases, your entire contact center environment is protected by NIST Level 5 post-quantum cryptography.

Phase 1 AI Protocol Discovery (2–4 Hours)

Discovery Phase	Duration	What Happens
Statistical Analysis	5–10 sec	Entropy, byte frequency, binary vs. text classification
ML Classification	10–20 sec	CNN + BiLSTM — 89%+ protocol family accuracy
Grammar Learning	1–2 min	PCFG inference + Transformer semantic learning
Parser Generation	30–60 sec	Auto-generate parsers at 50,000+ msg/sec
Adaptive Learning	Continuous	Error analysis and grammar refinement

Phase 2 Post-Quantum Encryption (1 Hour)

Subdomain	Algorithm	Latency	Use Case
Voice Signaling	ML-KEM-512 + Falcon-512	<50ms	SIP/SDP encryption
Voice Media (RTP)	ML-KEM-512 + Falcon-512	<20ms	Real-time media encryption
Payment Processing	ML-KEM-1024 + ML-DSA-87	<100ms	PCI-DSS Level 1
Call Recording	ML-KEM-1024 + ML-DSA-87	<1s	Quantum-safe long-term storage
Remote Agent Tunnel	ML-KEM-768 + ML-DSA-65	<200ms	VPN-less WFH access
Terminal Emulation	ML-KEM-768 + ML-DSA-65	<300ms	TN3270e/TN5250 sessions

Phase 3 Agentic AI Security — 78% Autonomous Response

QBITEL's agentic AI monitors all protected traffic continuously. 78% of security events are handled without human intervention. Every automated action generates a plain-language LLM-powered narrative — not an alert code. All LLM reasoning runs on-premise (Ollama/Llama 3) — no customer data leaves your network.

Threat Event	Autonomous Response	Time to Resolution
SIP injection attack	Block source, alert NOC, preserve evidence	<1 second
Terminal session hijacking	Terminate session, force re-auth	<2 seconds

Bulk customer data access	Rate-limit, supervisor notification	<5 seconds
Recording tampering	Integrity alert, evidence locked	<1 second
Rogue remote agent device	Quarantine endpoint, suspend sessions	<10 seconds
Toll fraud pattern detected	Block trunk, alert, forensics preserved	<1 second

3. CAPABILITY DEEP DIVES

Six purpose-built security modules for contact center environments

3.1 PCI-DSS Voice Compliance — Up to 80% Scope Reduction

Every system that touches cardholder data falls within PCI-DSS audit scope — recordings, agent desktops, CRM integrations. QBITEL eliminates cardholder data from your environment before it can be stored or accessed.

PCI Control	What QBITEL Does	Compliance Impact
DTMF Masking	Card digits clamped in headset AND recording (<5ms)	CHD never reaches agent or recording
Auto Pause/Resume	Recording pauses on payment detection automatically	Recording system exits PCI scope
PAN Detection	Real-time Luhn validation across all data streams	Accidental PAN caught immediately
Screen Masking	Last 4 digits only on agent desktop	Agent screen removed from scope
Recording Encryption	ML-KEM-1024 for archive	Safe against quantum decryption
Scope Tracking	Auto PCI scope per call/agent/tenant	Continuous evidence — not just at audit

3.2 Toll Fraud Prevention — 10 Pattern Types, <1 Second Detection

Fraud Type	Attack Pattern	QBITEL Response
IRSF	Premium-rate calls — 200+ country database	Block <1 second; forensics preserved
PBX Hacking	Unauthorized trunk access; off-hours spikes	Trunk quarantined; NOC alerted
Wangiri	Missed call callback to premium numbers	Pattern detected after 3 calls; blocked
Call Transfer Fraud	Transfer to premium-rate destinations	Whitelist validation on every transfer
Call Pumping	Artificially extended calls; revenue share abuse	Duration anomaly; call terminated
Bypass Fraud	SIM box termination; CLI manipulation	CLI anomaly detected; call blocked

3.3 Agent Desktop DLP — Six Exfiltration Channels Blocked

Threat Vector	Exfiltration Method	QBITEL Defense
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Clipboard	Copy-paste SSN/card numbers to personal files	Kernel-level PII clipboard block
Screen Capture	Screenshot customer data	Block PrintScreen + third-party tools
USB	Copy data to USB drive	USB storage blocked; events logged
Email/Chat	Email PII to personal accounts	Outbound PII pattern monitoring
Voice Reading	Read card numbers aloud during calls	Speech analytics detect CHD reading
Screen Scraping	Automated tools scraping agent desktop	eBPF-based scraping detection

Forensic Watermarking: Every agent screen carries an invisible watermark with agent ID and session timestamp — enabling post-incident attribution to the exact agent, session, and moment.

4. COMPLIANCE COVERAGE

Nine frameworks automated. Reports generated in under 10 minutes.

Framework	BPO Application	What QBITEL Automates
PCI-DSS 4.0	Voice payment processing	DTMF masking, recording encryption, agent controls, scope reduction
TCPA	Outbound calling	Consent tracking, DNC list enforcement, time-of-day restrictions
HIPAA	Healthcare BPO	PHI encryption, minimum necessary access, 6-year audit retention
SOC 2 Type II	Service organizations	Continuous monitoring, automated evidence, real-time alerting
GDPR	EU customer data	Recording consent, DSAR processing, retention/deletion automation
SOX	Financial services	Recording integrity, tamper-evident audit trails, 7-year retention
GLBA	Financial data	Data classification, access controls, breach notification
FCA/MiFID II	UK/EU financial recording	All-call recording, quantum-safe encryption, retention
NIST PQC	Quantum-safe transition	ML-KEM + ML-DSA across all voice and data channels

■ *Compliance reports generated on-demand in under 10 minutes per framework. Blockchain-backed audit trails for tamper evidence. Multi-tenant isolation ensures each enterprise client receives fully independent compliance reporting.*

5. REAL-WORLD SCENARIOS

Proven results across financial services, healthcare, and multi-tenant BPO environments

SCENARIO A > Financial Services BPO — 5,000 Seats

Challenge: Fortune 500 bank BPO handles credit card disputes over legacy Avaya infrastructure. PCI-DSS audit scope covers the entire contact center. Annual audit cost: \$2.3M.

QBITEL: Network tap in 30 min. AI discovers SIP + TN3270e in 2 hours. PQC-SRTP and DTMF masking active before end of business. Agents work uninterrupted. PCI scope reduced 80%. Annual audit cost reduced by \$1.7M.

SCENARIO B › Healthcare BPO — 2,000 Remote Agents

Challenge: Remote agents handle patient scheduling and insurance verification from home networks. HIPAA requires audit trails and PHI encryption — with zero visibility into agent home environments.

QBITEL: VPN-less quantum-safe tunnels deployed to 2,000 agents. Continuous endpoint compliance (WPA3, disk encryption, antivirus). PHI exfiltration monitoring across all channels. First HIPAA audit: zero findings.

SCENARIO C › Toll Fraud — The \$50,000 Weekend Attack

Challenge: Contact center discovers \$50,000 in fraudulent Caribbean premium-rate calls on Monday morning. Attack ran all weekend through one compromised SIP trunk.

With QBITEL: IRSF pattern detected after the 3rd call. Trunk isolated automatically. NOC alerted. Forensic evidence preserved. Total loss: \$200 vs. \$50,000 without QBITEL.

SCENARIO D › Multi-Tenant BPO — Banking, Healthcare, Retail

Challenge: One BPO facility, three enterprise clients with different compliance frameworks. Three separate annual audits. Separate infrastructure per client is cost-prohibitive.

QBITEL: Per-tenant cryptographic key isolation. Per-tenant compliance: PCI-DSS, HIPAA, SOC 2. Three independent compliance reports — generated on-demand in under 10 minutes each.

6. INTEGRATION

Works with everything you already have — nothing replaced

6.1 PBX & Telephony Platforms

Platform	Integration Method	Infrastructure Change
Avaya Aura / CM	TSAPI/DMCC with PQC tunnel	None
Cisco CUCM	CTI-OS / Finesse API + PQC	None
Genesys Cloud	REST API with PQC-TLS	None
Asterisk / FreePBX	AMI/ARI with PQC tunnel	None
Legacy PBX (any)	Protocol-level overlay	None — fully agnostic

6.2 Deployment — 4 Steps to Full Protection

Step	Duration	Activity
1. Network Tap	30 minutes	Non-invasive tap on voice/data network — passive observation
2. Protocol Discovery	2–4 hours	AI identifies all protocols, traffic patterns, data flows
3. Encryption Activation	1 hour	PQC activated for all discovered protocols automatically
4. Policy Deployment	30 minutes	BPO security policies configured and validated
Total	4–6 hours	Full quantum-safe protection. Zero downtime. Nothing replaced.

7. PERFORMANCE SPECIFICATIONS

Enterprise scale — zero compromise on voice quality

Metric	Performance	Business Impact
Voice PQC overhead	<2ms	Inaudible — within ITU-T G.114 budget
DTMF masking latency	<5ms	Callers cannot detect payment flow delay
Toll fraud detection	<1 second	Fraud stopped before meaningful loss
Concurrent agent sessions	20,000+	Enterprise scale from day one
Recording encryption	10,000+ streams	Entire recording estate simultaneously
Autonomous threat resolution	78% no-touch	SOC handles exceptions, not routine alerts
PAN detection	<50ms	Real-time Luhn validation on all streams

Compliance report generation	<10 minutes	On-demand for any framework, any client
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8. WHY QBITEL BRIDGE

The only platform purpose-built for BPO contact center security

8.1 vs. Traditional Security Vendors

Capability	Traditional Vendors	QBITEL Bridge
Legacy protocol support	Known protocols only	AI discovers unknown/undocumented protocols
Quantum cryptography	Not available	NIST Level 5 — ML-KEM-1024 + ML-DSA-87
Deployment	Weeks to months	4–6 hours, zero downtime
BPO-specific controls	Generic security policies	DTMF masking, toll fraud, DLP, multi-tenant
Threat response	Alert-based, SOC review	78% autonomous LLM reasoning
Compliance	Manual evidence collection	9 frameworks automated, <10 min reports
Air-gapped AI	Cloud-dependent	On-premise Ollama — no data egress

8.2 vs. PBX Replacement

Factor	PBX Replacement	QBITEL Bridge
Cost	\$5M+ hardware + migration + training	Fraction of replacement cost
Downtime	Weeks of migration risk	Zero — 4–6 hour deployment
Timeline	12–18 months to production	Full protection same business day
Quantum-readiness	Depends on vendor roadmap	NIST Level 5 from day one
Agent impact	Full retraining required	Zero — agents notice nothing

9. NEXT STEPS

Full quantum-safe protection in under one week

STEP 1 › Discovery Assessment — Free, 2 Hours

Passive network tap deployed in your environment.
Report delivered showing: which protocols are running, what is unencrypted,
and where your PCI/HIPAA compliance scope currently sits.
No commitment. No infrastructure changes.

STEP 2 › **Proof of Concept — 2 Weeks**

Full QBITEL Bridge deployment against your production traffic.
Real threats. Live compliance reporting. Toll fraud monitoring.
ROI measured against your current state baseline — before you sign.

STEP 3 › **Production Deployment — 4–6 Hours**

Zero-downtime deployment. All protocols protected. All agents covered.
All 9 compliance frameworks active. Full documentation, runbooks,
and dedicated support for the first 90 days.

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Email	Website	Schedule a call

QBITEL Bridge — Because the Human API Deserves Quantum-Safe Protection.

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